

Package ‘manMetaVAR’

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Title Multivariate Meta-Analysis of Vector Autoregressive Model
Coefficients

Version 0.9.1

Description Research compendium for the manuscript
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Multivariate Meta-Analysis of Vector Autoregressive Model Coefficients.
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BugReports <https://github.com/jeksterslab/manMetaVAR/issues>

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Compress	<i>Compress Replication</i>
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Description

Compress Replication

Usage

Compress(taskid, repid, output_folder)

Arguments

- taskid Positive integer. Task ID.
- repid Positive integer. Replication ID.
- output_folder Character string. Output folder.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

FitDTVAR*Fit the Model using the fitDTVARMx Package*

Description

The function fits the model using the [fitDTVARMx](#) package.

Usage

```
FitDTVAR(data, ncores = NULL)
```

Arguments

<code>data</code>	R object. Output of the GenData() function.
<code>ncores</code>	Positive integer. Number of cores to use.

See Also

Other Model Fitting Functions: [FitJAGS\(\)](#), [FitMLVAR\(\)](#)

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitDTVAR(
  data = data,
  ncores = parallel::detectCores()
)
summary(fit)

## End(Not run)
```

FitJAGS*Fit the Model using JAGS*

Description

The function fits the model using JAGS.

Usage

```
FitJAGS(
  data,
  n_chains = 4,
  n_adapt = 10000,
  n_iter = 10000,
  thin = 1,
  seed = NULL
)
```

Arguments

<code>data</code>	R object. Output of the GenData() function.
<code>n_chains</code>	The number of parallel chains for the model.
<code>n_adapt</code>	The number of iterations for adaptation.
<code>n_iter</code>	Number of iterations to monitor.
<code>thin</code>	Thinning interval for monitors.
<code>seed</code>	Integer. Random seed.

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMLVAR\(\)](#)

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitJAGS(data = data)
summary(fit)

## End(Not run)
```

FitMetaVAR

Multivariate Meta-Analysis using the metaVAR Package

Description

The function performs multivariate meta-analysis using the [metaVAR](#) package.

Usage

```
FitMetaVAR(fit, ncores = NULL)
```

Arguments

fit	R object. Output of the FitDTVARmX() function.
ncores	Positive integer. Number of cores to use.

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitDTVAR(
  data = data,
  ncores = parallel::detectCores()
)
pooled <- FitMetaVAR(
  fit = fit,
  ncores = parallel::detectCores()
)
summary(pooled)

## End(Not run)
```

FitMLVAR

*Fit the Model using the mlVAR Package***Description**

The function fits the model using the [mlVAR](#) package.

Usage

```
FitMLVAR(data, ncores = NULL)
```

Arguments

data	R object. Output of the GenData() function.
ncores	Positive integer. Number of cores to use.

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitJAGS\(\)](#)

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitMLVAR(data = data)
summary(fit)

## End(Not run)
```

GenData

Simulate Data

Description

The function simulates data using the `simStateSpace::SimSSMIVary()` function.

Usage

```
GenData(taskid)
```

Arguments

taskid Positive integer. Task ID.

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)

## End(Not run)
```

params

Simulation Parameters

Description

Simulation Parameters

Usage

```
params
```

Format

A dataframe with 25 rows and 3 columns:

taskid Simulation Task ID.

n Sample size.

time Number of measurement occasions.

Author(s)

Ivan Jacob Agaloos Pesigan

plot.manmetavar_data *Plot Method for an Object of Class manmetavar_data*

Description

Plot Method for an Object of Class manmetavar_data

Usage

```
## S3 method for class 'manmetavar_data'  
plot(x, ...)
```

Arguments

x Object of class manmetavar_data.
... additional arguments.

Author(s)

Ivan Jacob Agaloos Pesigan

Sim *Simulation Replication*

Description

Simulation Replication

Usage

```
Sim(  
  taskid,  
  repid,  
  output_folder,  
  overwrite,  
  integrity,  
  seed,  
  n_chains,  
  n_adapt,  
  n_iter,  
  thin,  
  run_jags  
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
seed	Integer. Random seed.
n_chains	The number of parallel chains for the model.
n_adapt	The number of iterations for adaptation.
n_iter	Number of iterations to monitor.
thin	Thinning interval for monitors.
run_jags	Logical. If run_jags = TRUE, run JAGS model.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitDTVAR

Simulation Replication - FitDTVAR

Description

Simulation Replication - FitDTVAR

Usage

```
SimFitDTVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitJAGS

Simulation Replication - FitJAGS

Description

Simulation Replication - FitJAGS

Usage

```
SimFitJAGS(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  n_chains,
  n_adapt,
  n_iter,
  thin
)
```

Arguments

<code>taskid</code>	Positive integer. Task ID.
<code>repid</code>	Positive integer. Replication ID.
<code>output_folder</code>	Character string. Output folder.
<code>seed</code>	Integer. Random seed.
<code>suffix</code>	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
<code>overwrite</code>	Logical. Overwrite existing output in <code>output_folder</code> .
<code>integrity</code>	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .
<code>n_chains</code>	The number of parallel chains for the model.

n_adapt	The number of iterations for adaptation.
n_iter	Number of iterations to monitor.
thin	Thinning interval for monitors.

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMetaVAR	<i>Simulation Replication - FitMetaVAR</i>
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Description

Simulation Replication - FitMetaVAR

Usage

SimFitMetaVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMLVAR	<i>Simulation Replication - FitMLVAR</i>
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Description

Simulation Replication - FitMLVAR

Usage

SimFitMLVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFN	<i>Simulation File Name</i>
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Description

Simulation File Name

Usage

```
SimFN(output_type, output_folder, suffix)
```

Arguments

output_type	Character string. Output type. Valid values include "data", "fit-dt-var-mx", "fit-meta-var-mx", "fit-ml-var", and "fit-jags".
output_folder	Character string. Output folder.
suffix	Character string. Output of <code>manCTMed::SimSuffix()</code> .

Value

Returns a character string file name with the `output_folder` in the OS-specific format.

SimGenData	<i>Simulation Replication - GenData</i>
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Description

Simulation Replication - GenData

Usage

```
SimGenData(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimProj	<i>Simulation Project Name</i>
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Description

Simulation Project Name

Usage

SimProj()

Value

Returns the project name as a character string.

Author(s)

Ivan Jacob Agaloos Pesigan

summary.manmetavar_data	<i>Summary Method for an Object of Class manmetavar_data</i>
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Description

Summary Method for an Object of Class manmetavar_data

Usage

```
## S3 method for class 'manmetavar_data'  
summary(object, ...)
```

Arguments

object	Object of class manmetavar_data.
...	additional arguments.

Author(s)

Ivan Jacob Agaloos Pesigan

summary.manmetavar_fitdtvar

Summary Method for an Object of Class manmetavar_fitdtvar

Description

Summary Method for an Object of Class manmetavar_fitdtvar

Usage

```
## S3 method for class 'manmetavar_fitdtvar'  
summary(object, ...)
```

Arguments

object	Object of class manmetavar_fitdtvar.
...	additional arguments.

Author(s)

Ivan Jacob Agaloos Pesigan

summary.manmetavar_fitjags

Summary Method for an Object of Class manmetavar_fitjags

Description

Summary Method for an Object of Class manmetavar_fitjags

Usage

```
## S3 method for class 'manmetavar_fitjags'  
summary(object, ...)
```

Arguments

object	Object of class manmetavar_fitjags.
...	additional arguments.

Author(s)

Ivan Jacob Agaloos Pesigan

`summary.manmetavar_fitmetavar`*Summary Method for an Object of Class manmetavar_fitmetavar*

Description

Summary Method for an Object of Class manmetavar_fitmetavar

Usage

```
## S3 method for class 'manmetavar_fitmetavar'  
summary(object, ...)
```

Arguments

object	Object of class manmetavar_fitmetavar.
...	additional arguments.

Author(s)

Ivan Jacob Agaloos Pesigan

`summary.manmetavar_fitmlvar`*Summary Method for an Object of Class manmetavar_fitmlvar*

Description

Summary Method for an Object of Class manmetavar_fitmlvar

Usage

```
## S3 method for class 'manmetavar_fitmlvar'  
summary(object, ...)
```

Arguments

object	Object of class manmetavar_fitmlvar.
...	additional arguments.

Author(s)

Ivan Jacob Agaloos Pesigan

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